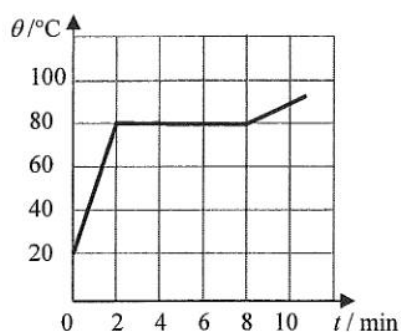


2012 Q4

4. Which of the following descriptions is correct ?
- A. When water at 25°C is heated to 50°C, both the kinetic energy and potential energy of the water molecules increase.
 - B. When water at 25°C is heated to 50°C, only the potential energy of the water molecules increases.
 - C. When water boils at 100°C and turns into steam, the kinetic energy of the water molecules increases.
 - D. When water boils at 100°C and turns into steam, the potential energy of the water molecules increases.

2014 Q2

2.

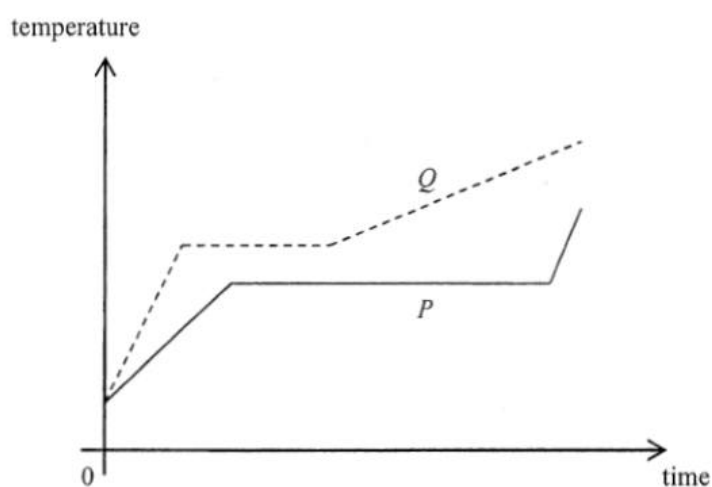


An electric heater of constant power is used to heat a solid substance X which is insulated from the surroundings. The variation of its temperature θ with time t is shown above. X has a specific heat capacity of $800 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ in its solid state. What is the specific latent heat of fusion of X ?

- A. 144 kJ kg^{-1}
- B. 192 kJ kg^{-1}
- C. 202 kJ kg^{-1}
- D. Answer cannot be found as both the mass of X and the power of the heater are not known.

2017 Q2

2. Same mass of solids P and Q are heated at the same rate. The temperature-time graphs of the two substances are shown below.



Which of the following comparisons about their melting points and specific latent heats of fusion is correct ?

	higher melting point	larger specific latent heat of fusion
A.	P	P
B.	P	Q
C.	Q	P
D.	Q	Q

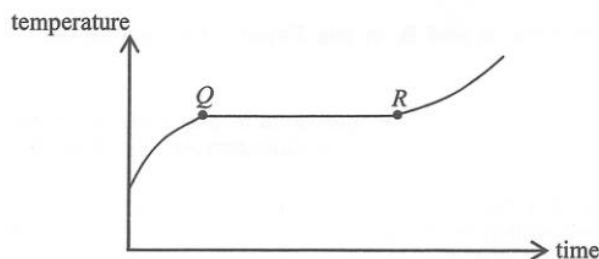
2017 Q3

3. Which of the following statements about the internal energy of a substance are correct ?

- (1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
 - (2) When a vapour condenses, its internal energy decreases.
 - (3) When a liquid evaporates, the internal energy of the remaining liquid increases.
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

2019 Q2

2.

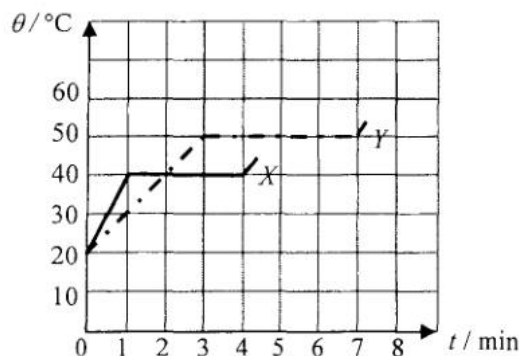


A substance undergoes the fusion process. The figure shows how the substance's temperature varies with time. Its temperature remains constant during the period from Q to R . Which of the following deductions **within this period** is/are correct?

- (1) The substance does not absorb heat.
 - (2) The mass ratio of the substance in solid and liquid states remains constant throughout.
 - (3) The average potential energy of the molecules of the substance increases with time.
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

2021 Q1

1. Solid substances X and Y of equal mass are heated by heaters of power $2P$ and P respectively. The graph shows how the temperature θ of each substance varies with the heating time t .



What is the ratio of the specific latent heat of fusion of X to that of Y ?

- A. 3 : 2
 B. 3 : 4
 C. 4 : 3
 D. 2 : 3

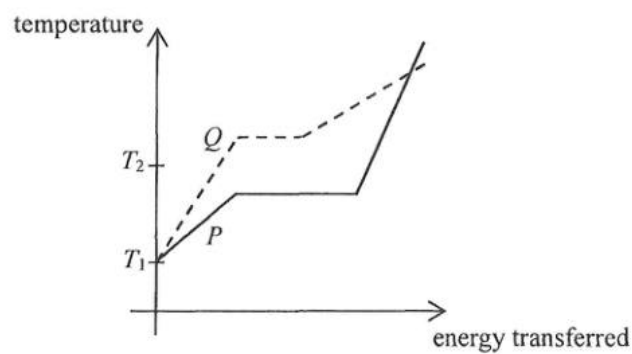
2022 Q2

2. When ice melts into water at 0°C , what happens to the molecules during the melting process?

- (1) their average separation increases
 - (2) their average kinetic energy increases
 - (3) their average potential energy increases
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

2024 Q1

1. Substance P and Q are of the same mass and both are solids at temperature T_1 . The graph below shows the variation of temperature of the substances with energy transferred.



Which statement below is correct ?

- A. P has a higher melting point than Q .
- B. The specific latent heat of fusion of P is smaller than that of Q .
- C. The specific heat capacity of solid P is greater than that of solid Q .
- D. At temperature T_2 , P is gas while Q is solid.